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Linear Algebra I (MA2102)

Mid Sem Exam

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Full marks: 20. Write your answers as neatly as possible. No points will be awarded for the answers which lack proper justification.

Exercise 1. (3 + 3 + 3 points)
Let $P_2(\mathbb{R})$ denote the polynomials of degree less than or equal to 2, having real coefficients and let $\mathbb{B} := (2x^2 + 1, 3x + 4x^2, 5 + 6x^2) \subset P_2(\mathbb{R})$.
(i) Justify whether $\mathbb{B}$ is a basis of $P_2(\mathbb{R})$.
(ii) Write down the matrix of the linear operator

$$\frac{d}{dx} : P_2(\mathbb{R}) \to P_2(\mathbb{R})$$

with respect to $\mathbb{B}$.
(iii) Compute the matrix of change of basis from $\mathbb{B}$ to $\mathbb{B}' := (1, 1 + x, 1 + x + x^2)$.

Exercise 2. (2 + 3 points)
Let $A \in \mathrm{GL}_n(\mathbb{R})$, let $V := M_n(\mathbb{R})$ and let the linear operator $T : V \to V$ be defined by

$$T(v) = Av - vA \quad \text{for all } v \in V.$$

(i) Find an eigenvector of T.
(ii) Prove or disprove: $\mathrm{rank}(T) = n^2$.

Exercise 3. (6 points)
Show that an $n \times n$ real matrix is invertible if and only if its rows span $\mathbb{R}^n$.